BANG PHAN CONG NHIEM VU & HUONG DAN TRIEN KHAI DO AN

DO AN LAP TRINH SOCKET: P2P LAN CHAT & FILE SHARING

Repository: https://github.com/VN-DEV/HCMUS-Socket-Project

1. THANH VIEN NHOM VA TY TRONG CONG VIEC

STT	Ho va ten	Vai tro	Pham vi phu trach chinh
1	**Nguyen Manh Tri**	Nhom truong	Kien truc tong the, Protocol, UDP Discovery & Status, Dieu phoi he thong
2	**Do Duy Truong**	Thanh vien	TCP Server (Acceptor), TCP Connection, Nhan tin 1-1 thoi gian thuc
3	**Mai Huu Tri**	Thanh vien	Truyen file da luong P2P (Chunking 64KB, %), ChatManager & Giao dien Qt GUI

2. PHAN CHIA NHIEM VU KY THUAT CHI TIET

Thanh vien	File ma nguon	Nhiem vu cu the can hien thuc	Diem tieu chi
Nguyen Manh Tri (Nhom truong)	src/models/Peer.h src/protocol/Protocol.* src/network/UdpDiscovery.* src/main.cpp	• Xay dung cau truc du lieu Peer, ChatMessage. • Hien thuc serialize/parse cac goi tin: HELLO, AUTH, MSG, FILE, BYE. • Lap trinh UDP Broadcaster phat HELLO moi 3s qua cong 9999 (SO_BROADCAST). • Lap trinh UDP Listener nghe cong 9999 (SO_REUSEADDR) cap nhat online list. • Xu ly timeout 10s xoa peer offline; dong bo trang thai online/away.	35% (15d UDP + 10d Protocol + 10d Base)
Do Duy Truong	src/network/TcpServer.* src/network/TcpConnection.*	• Mo cong TCP lang nghe ket noi, luong acceptLoop don client moi. • Tu dong chon cong tiep theo (5001, 5002...) neu cong 5000 bi chiem. • Chu dong ket noi TCP (connect) toi IP:Port cua peer khi bat dau chat. • Gui bat tay AUTH|name|port| ngay sau khi ket noi thanh cong. • Luong nhan du lieu receiverLoop doc byte stream theo dung do dai length. • Dong goi va gui tin nhan MSG|length|text dam bao khong dính goi.	30% (20d TCP Chat + 10d Error Handling)
Mai Huu Tri	src/network/TcpConnection.cpp src/chat/ChatManager.* src/ui/MainWindow.*	• Hien thuc truyen file tren luong rieng (sendFileAsync) khong lam nghen chat. • Doc file theo chunk 64KB, gui FILE header va phat tin hieu tien trinh %. • Nhan file ghi vao downloads/, xoa file do dang neu dut ket noi. • Quan ly danh ba ket noi, lich su chat, dem so tin chua doc (unreadCount). • Hoan thien giao dien do hoa Qt: danh sach peer, khung chat, nut gui file.	35% (15d File + 15d UI + 5d Multi-peer)

3. QUY CHUAN GIAO THUC TANG UNG DUNG (APPLICATION FRAMING)

• UDP Broadcast: HELLO|<name>|<tcp_port>|<status> (Vi du: HELLO|Alice|5001|online)
• TCP Handshake: AUTH|<name>|<tcp_port>| (Gui ngay khi vua mo ket noi TCP thanh cong)
• TCP Message: MSG|<length>|<utf-8 text> (Vi du: MSG|5|Hello)

• TCP File Header: FILE|<filename>|<size_in_bytes>| (Theo sau la cac khoi nhi phan 64KB)
• TCP Disconnect: BYE| (Thong bao ngat ket noi an toan)

4. HUONG DAN BIEN DICH VA CHAY THU NGHIEM

```
# 1. Bien dich toan bo du an bang Ninja
cmake -B build -G "Ninja" -DCMAKE_EXPORT_COMPILE_COMMANDS=ON
cmake --build build

# 2. Khoi chay 2 may ao (hoac 2 cua so tren 1 may tinh de kiem tra P2P)
# Terminal 1: .\build\lan_chat.exe --name Alice --port 5001
# Terminal 2: .\build\lan_chat.exe --name Bob --port 5002
```